

1202SCG Calculus 1
1020ENG Engineering Mathematics 2

Week 6 Lecture

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Mid-Trimester Examination

- ▶ Duration: **80 minutes**.
- ▶ The examination is **closed book**.
- ▶ A scientific calculator is permitted.
- ▶ Symbolic algebra or calculus calculators are not permitted.
- ▶ See Learning@Griffith and emails from Jason for more information.

Examination Day

- ▶ When arriving at the venue, please wait outside. You will then be given directions on where to sit.
- ▶ Please arrive slightly before the scheduled lecture time:
 - ▶ Gold Coast: 4 pm
 - ▶ Nathan: 5 pm
- ▶ Remember your Student ID.
- ▶ Due to the mid-trimester examination, there will be no lecture in Week 7. Lectures will continue as normal in Week 8.

Related Rates of Change

Consider a real-world system where a quantity y depends on another quantity x . As x changes over time, y changes in response. Their rates of change are connected through the chain rule.

$$\frac{dy}{dt} = \frac{dy}{dx} \frac{dx}{dt}.$$

A Related-Rates Strategy

1. Define the changing variables and their units.
2. Write an equation relating the variables.
3. Differentiate the equation with respect to time.
4. Substitute the values for the instant being considered.
5. Solve for the required rate and interpret its sign.

Differentiate before substituting numerical values.

Related Rates: An Expanding Circle

The radius of a circular oil spill is increasing at 0.3 m/min.
How quickly is its area increasing when the radius is 5 m?

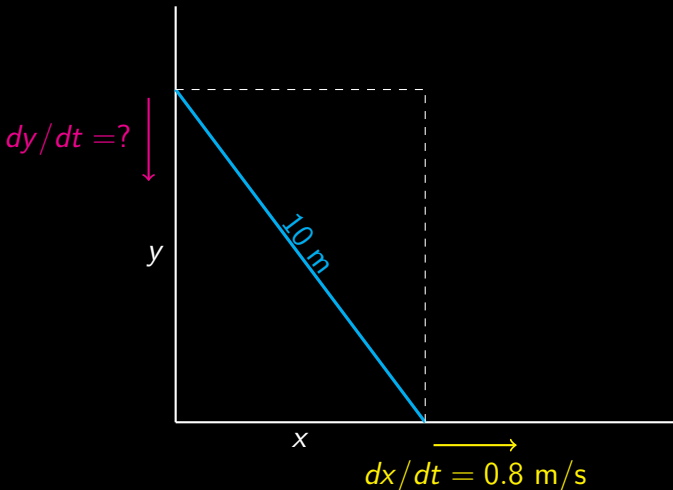
$$A = \pi r^2,$$
$$\frac{dA}{dt} = 2\pi r \frac{dr}{dt}.$$

At $r = 5$ and $dr/dt = 0.3$,

$$\begin{aligned}\frac{dA}{dt} &= 2\pi(5)(0.3) \\ &= 3\pi \text{ m}^2/\text{min}.\end{aligned}$$

Related Rates: A Sliding Ladder

A 10 m ladder rests against a vertical wall. Its base moves away from the wall at 0.8 m/s. How quickly is the top of the ladder moving when the base is 6 m from the wall?



Sliding Ladder: Solution

The geometry gives

$$x^2 + y^2 = 100.$$

When $x = 6$, $y = 8$. Differentiate with respect to time:

$$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0,$$

$$6(0.8) + 8 \frac{dy}{dt} = 0,$$

$$\frac{dy}{dt} = -0.6 \text{ m/s}.$$

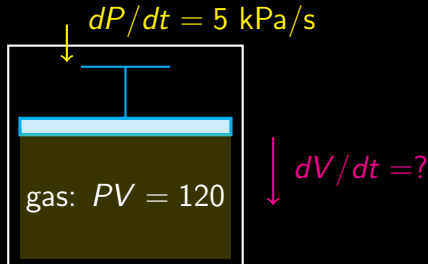
The negative sign means the top of the ladder is moving downward at 0.6 m/s.

Related Rates: Gas Under Compression

A gas is maintained at constant temperature, so its pressure P and volume V satisfy

$$PV = 120.$$

The pressure is increasing at 5 kPa/s. Find dV/dt when $P = 30$ kPa.



Gas Under Compression: Solution

At this instant, $V = 120/30 = 4$ L. Differentiating with respect to time gives

$$\begin{aligned}P \frac{dV}{dt} + V \frac{dP}{dt} &= 0, \\30 \frac{dV}{dt} + 4(5) &= 0, \\ \frac{dV}{dt} &= -\frac{2}{3} \text{ L/s}.\end{aligned}$$

The gas volume is decreasing at $2/3$ L/s.

Related Rates: Electrical Power

A resistor has constant resistance $R = 6 \, \Omega$. The voltage across it is increasing at $0.5 \, \text{V/s}$. Find the rate at which the electrical power changes when the voltage is $12 \, \text{V}$.

$$P = \frac{V^2}{R}.$$

Since R is constant,

$$\begin{aligned}\frac{dP}{dt} &= \frac{2V}{R} \frac{dV}{dt} \\ &= \frac{2(12)}{6} (0.5) \\ &= 2 \, \text{W/s}.\end{aligned}$$

The electrical power is increasing at $2 \, \text{W/s}$.

Indeterminate Forms in Limits

Direct substitution sometimes produces an expression whose value is not yet determined.

These are not answers. They indicate that more analysis is required.

The two forms used with L'Hôpital's rule are

$$\frac{0}{0} \quad \text{and} \quad \frac{\pm\infty}{\pm\infty}.$$

L'Hôpital's Rule

If direct substitution gives $0/0$ or ∞/∞ , then under suitable differentiability conditions,

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}.$$

Differentiate the numerator and denominator separately.

$$\frac{d}{dx} \left(\frac{f}{g} \right) \neq \frac{f'}{g'}.$$

Using L'Hôpital's Rule

1. Substitute into the original limit.
2. Confirm that the form is $0/0$ or ∞/∞ .
3. Differentiate the numerator and denominator separately.
4. Evaluate the new limit.
5. If it is still indeterminate, the rule may be applied again.

Do not use the rule before checking the indeterminate form.

L'Hôpital's Rule: Example 1

Evaluate

$$\lim_{x \rightarrow 0} \frac{e^{2x} - 1}{x}.$$

Direct substitution gives 0/0, so

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{e^{2x} - 1}{x} &= \lim_{x \rightarrow 0} \frac{2e^{2x}}{1} \\ &= 2. \end{aligned}$$

L'Hôpital's Rule: Repeated Use

Evaluate

$$\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}.$$

Substitution gives 0/0:

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2} &= \lim_{x \rightarrow 0} \frac{e^x - 1}{2x} \\ &= \lim_{x \rightarrow 0} \frac{e^x}{2} \\ &= \frac{1}{2}.\end{aligned}$$

L'Hôpital's Rule at Infinity

Evaluate

$$\lim_{x \rightarrow \infty} \frac{3x^2 + 5x - 1}{x^2 + 4}.$$

The form is ∞/∞ :

$$\begin{aligned}\lim_{x \rightarrow \infty} \frac{3x^2 + 5x - 1}{x^2 + 4} &= \lim_{x \rightarrow \infty} \frac{6x + 5}{2x} \\ &= \lim_{x \rightarrow \infty} \frac{6}{2} \\ &= 3.\end{aligned}$$

Rewriting Other Indeterminate Forms

L'Hôpital's rule applies to quotients. Other forms must first be rewritten.

For example,

$$\lim_{x \rightarrow 0^+} x \ln x$$

has the form $0 \cdot (-\infty)$. Rewrite it as

$$\begin{aligned}\lim_{x \rightarrow 0^+} \frac{\ln x}{1/x} &= \lim_{x \rightarrow 0^+} \frac{1/x}{-1/x^2} \\ &= \lim_{x \rightarrow 0^+} (-x) = 0.\end{aligned}$$

When L'Hôpital's Rule Does Not Apply

Consider

$$\lim_{x \rightarrow 1} \frac{x+2}{x-1}.$$

Direct substitution gives $3/0$. This is *not* an indeterminate form, so L'Hôpital's rule cannot be used.

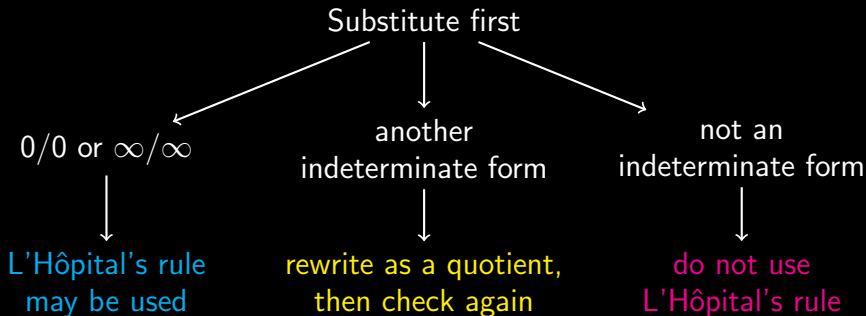
In fact,

$$\lim_{x \rightarrow 1^-} \frac{x+2}{x-1} = -\infty, \quad \lim_{x \rightarrow 1^+} \frac{x+2}{x-1} = \infty.$$

Therefore, the two-sided limit does not exist.

Can L'Hôpital's Rule Be Used?

Begin by substituting the limiting value into the quotient.



Taylor Series

A Taylor series represents a smooth function near $x = a$ using a polynomial.

The polynomial is chosen to match the function's

- ▶ value,
- ▶ gradient,
- ▶ concavity, and
- ▶ higher derivatives at $x = a$.

Close to a , a complicated function can behave like a simple polynomial.

Factorials

$$1! = 1,$$

$$2! = 2 \times 1 = 2,$$

$$3! = 3 \times 2 \times 1 = 6,$$

$$4! = 4 \times 3 \times 2 \times 1 = 24,$$

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120,$$

$$6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720.$$

$$0! = 1$$

Taylor Series About $x = a$

The Taylor series of $f(x)$ about $x = a$ can be written in sigma notation as

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n.$$

Expanding the first few terms gives

$$\begin{aligned} f(x) = & f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 \\ & + \frac{f'''(a)}{3!}(x - a)^3 + \cdots . \end{aligned}$$

Maclaurin Series

A Taylor series about $x = 0$ is called a **Maclaurin series**:

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots$$

Common examples are

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots,$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots,$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

Taylor Series: Sine Example

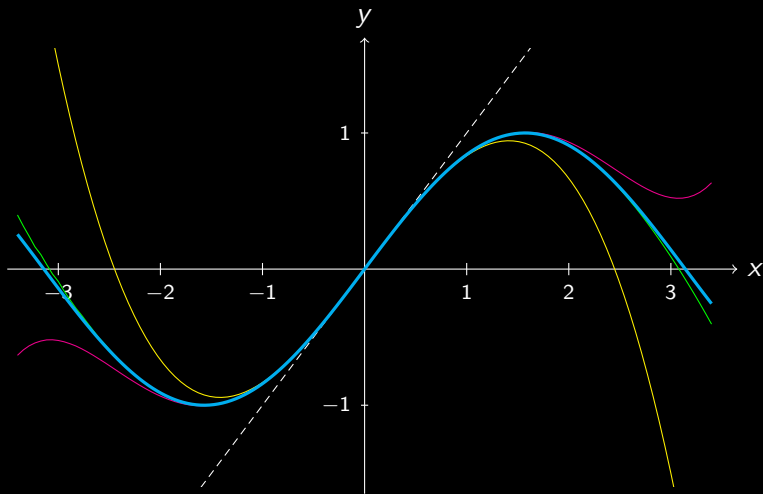
Find the Taylor series of $f(x) = \sin x$ about $x = 0$, up to the x^3 term.

$$\begin{array}{ll} f(x) = \sin x, & f(0) = 0, \\ f'(x) = \cos x, & f'(0) = 1, \\ f''(x) = -\sin x, & f''(0) = 0, \\ f'''(x) = -\cos x, & f'''(0) = -1. \end{array}$$

Therefore,

$$\begin{aligned} \sin x &= 0 + x + \frac{0}{2!}x^2 + \frac{-1}{3!}x^3 + \cdots \\ &= x - \frac{x^3}{6} + \cdots . \end{aligned}$$

Taylor Polynomials for $\sin x$



$\sin x$

$P_1 = x$

$P_3 = x - x^3/3!$

$P_5 = P_3 + x^5/5!$

$P_7 = P_5 - x^7/7!$

Taylor Series: Exponential Example

Find the Taylor series of $f(x) = 2e^{3x}$ about $x = 0$, up to the x^3 term.

$$\begin{array}{ll} f(x) = 2e^{3x}, & f(0) = 2, \\ f'(x) = 6e^{3x}, & f'(0) = 6, \\ f''(x) = 18e^{3x}, & f''(0) = 18, \\ f'''(x) = 54e^{3x}, & f'''(0) = 54. \end{array}$$

Hence,

$$\begin{aligned} 2e^{3x} &= 2 + 6x + \frac{18}{2!}x^2 + \frac{54}{3!}x^3 + \cdots \\ &= 2 + 6x + 9x^2 + 9x^3 + \cdots. \end{aligned}$$

Taylor Series and Limits

Taylor series can reveal the first nonzero term in a limit.

Evaluate without L'Hôpital's rule:

$$\lim_{x \rightarrow 0} \frac{2e^{3x} - 2 - 6x - 9x^2}{x^3}.$$

Since

$$2e^{3x} = 2 + 6x + 9x^2 + 9x^3 + \cdots,$$

the limit becomes

$$\lim_{x \rightarrow 0} \frac{9x^3 + \cdots}{x^3} = 9.$$

Taylor Series About a Nonzero Point

Find the Taylor series of $f(x) = \ln x$ about $x = 1$ up to the cubic term.

$$f(1) = 0,$$

$$f'(x) = \frac{1}{x},$$

$$f'(1) = 1,$$

$$f''(x) = -\frac{1}{x^2},$$

$$f''(1) = -1,$$

$$f'''(x) = \frac{2}{x^3},$$

$$f'''(1) = 2.$$

Hence,

$$\ln x = (x - 1) - \frac{(x - 1)^2}{2} + \frac{(x - 1)^3}{3} + \cdots .$$

Taylor Series: Square-Root Example

Find the Taylor series of $f(x) = \sqrt{x}$ about $x = 4$, up to the cubic term.

$$\begin{array}{ll} f(x) = \sqrt{x}, & f(4) = 2, \\ f'(x) = \frac{1}{2\sqrt{x}}, & f'(4) = \frac{1}{4}, \\ f''(x) = -\frac{1}{4x^{3/2}}, & f''(4) = -\frac{1}{32}, \\ f'''(x) = \frac{3}{8x^{5/2}}, & f'''(4) = \frac{3}{256}. \end{array}$$

Hence,

$$\sqrt{x} = 2 + \frac{1}{4}(x - 4) - \frac{1}{64}(x - 4)^2 + \frac{1}{512}(x - 4)^3 + \dots$$

Week 6 Summary

- ▶ Related rates use the chain rule to connect changing quantities.
- ▶ L'Hôpital's rule applies only to $0/0$ or ∞/∞ forms.
- ▶ Other indeterminate forms must be rewritten before applying the rule.
- ▶ Taylor series match a function and its derivatives at an expansion point.
- ▶ Taylor series provide local approximations and can evaluate limits.